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Zhang et al.

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(54) **LIGHT SOURCE MODULE**

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F21V 5/04 (2006.01)
F21Y 113/10 (2016.01)

(52) **U.S. Cl.**

CPC ***F21V 5/02*** (2013.01); ***F21V 5/04*** (2013.01); *F21Y 2113/10* (2016.08)

(58) **Field of Classification Search**

CPC F21Y 2113/10
See application file for complete search history.

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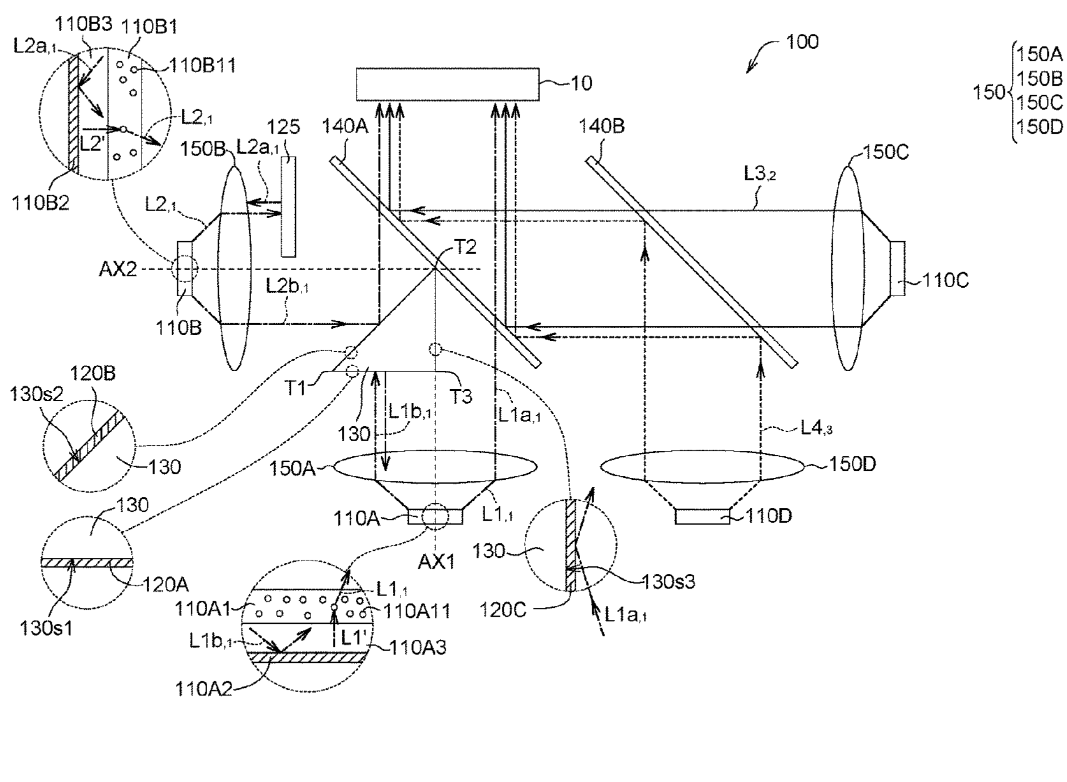
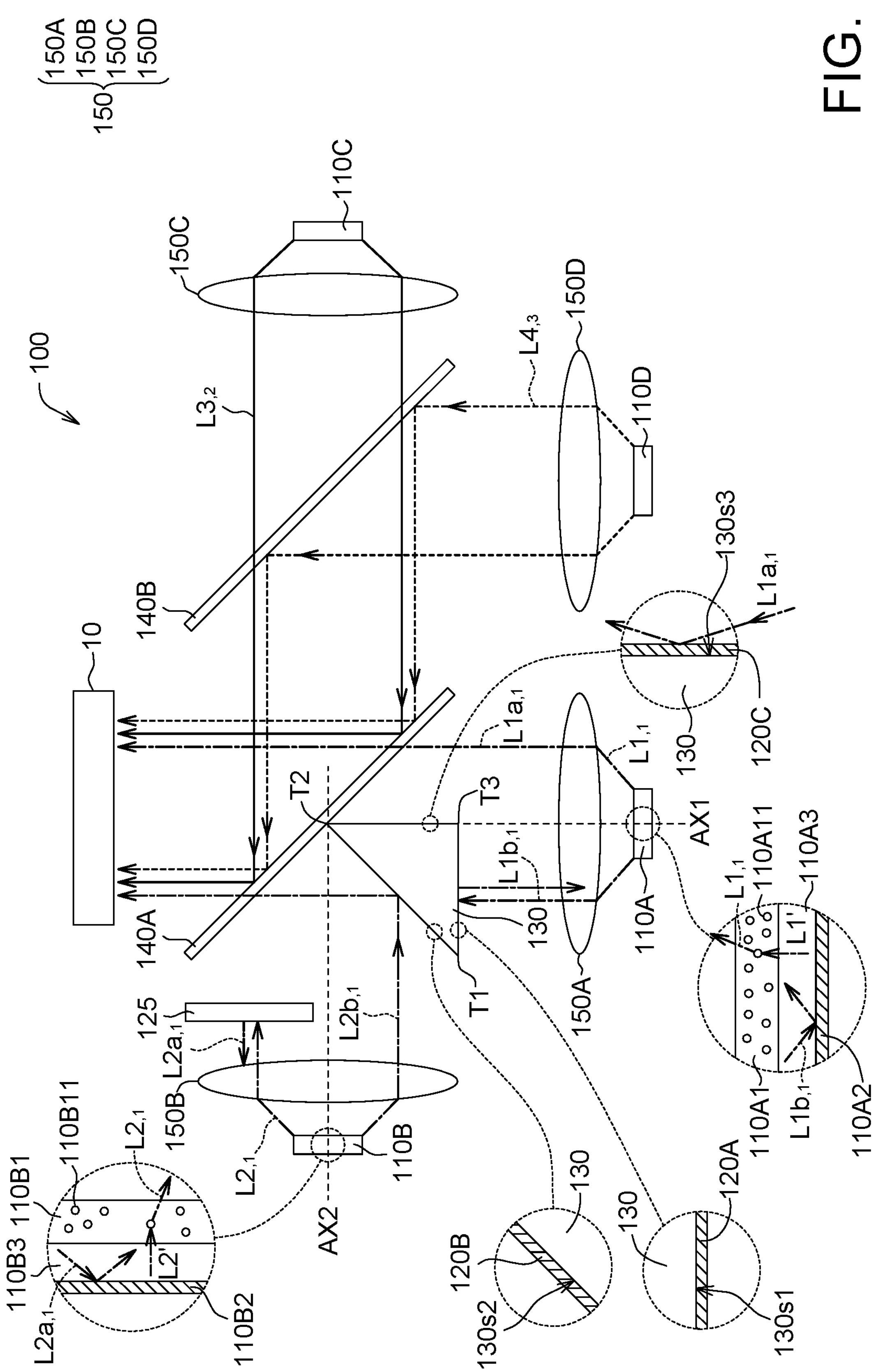
Primary Examiner — Fatima N Farokhrooz

(57) **ABSTRACT**

A light source module includes a first light source, a second light source, a first reflective layer and a second reflective layer. The first light source is configured to emit a first light having a first wavelength. The second light source is configured to emit a second light having the first wavelength. The first reflective layer is disposed opposite to the first light source and configured to reflect the first light. The second reflective layer is disposed opposite to the second light source and configured to reflect the second light. The first reflective layer and the second reflective layer are configured to reflect light of any wavelength in a visible spectrum.

7 Claims, 4 Drawing Sheets




100
150
150A
150B
150C
150D
110C
110D
110A
110B
10
140A
140B
125
130
120A
120B
120C
130s1
130s2
130s3
T1
T2
T3
AX1
AX2
L3,2
L4,3
L1,1
L1a,1
L1b,1
L2,1
L2a,1
L2b,1
L1'
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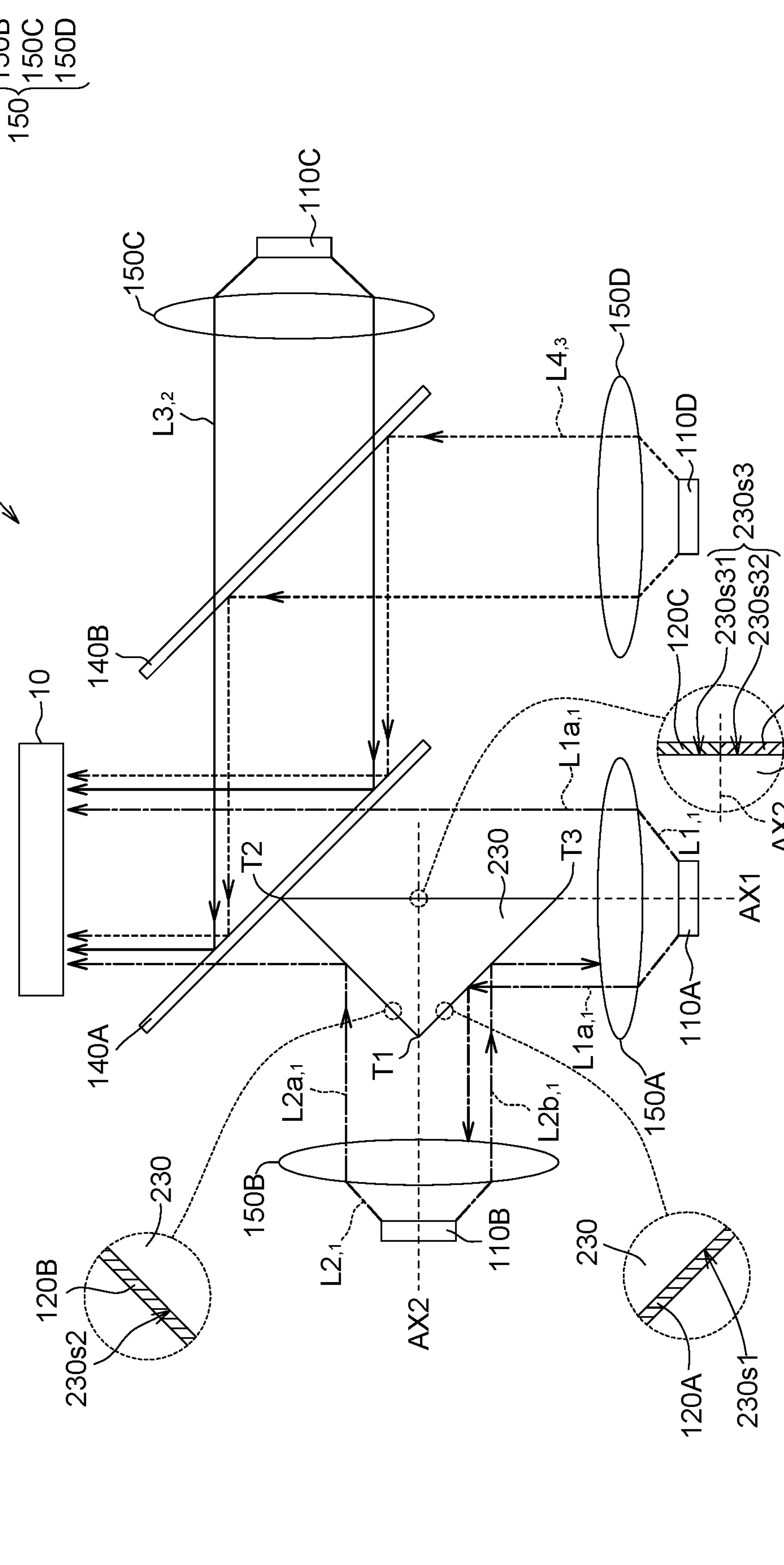
110A1
110A11
110A2
110A3
110B1
110B11
110B2
110B3

FIG. 1



150
150A
150B
150C
150D
200
110C
150C
150D
L3,2
L4,3
110D
230s3
230s31
230s32
220A
120C
140B
10
L1a,1
230
AX2
L1,1
T2
230
T3
AX1
110A
140A
T1
L1a,1
150A
L2a,1
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AX2
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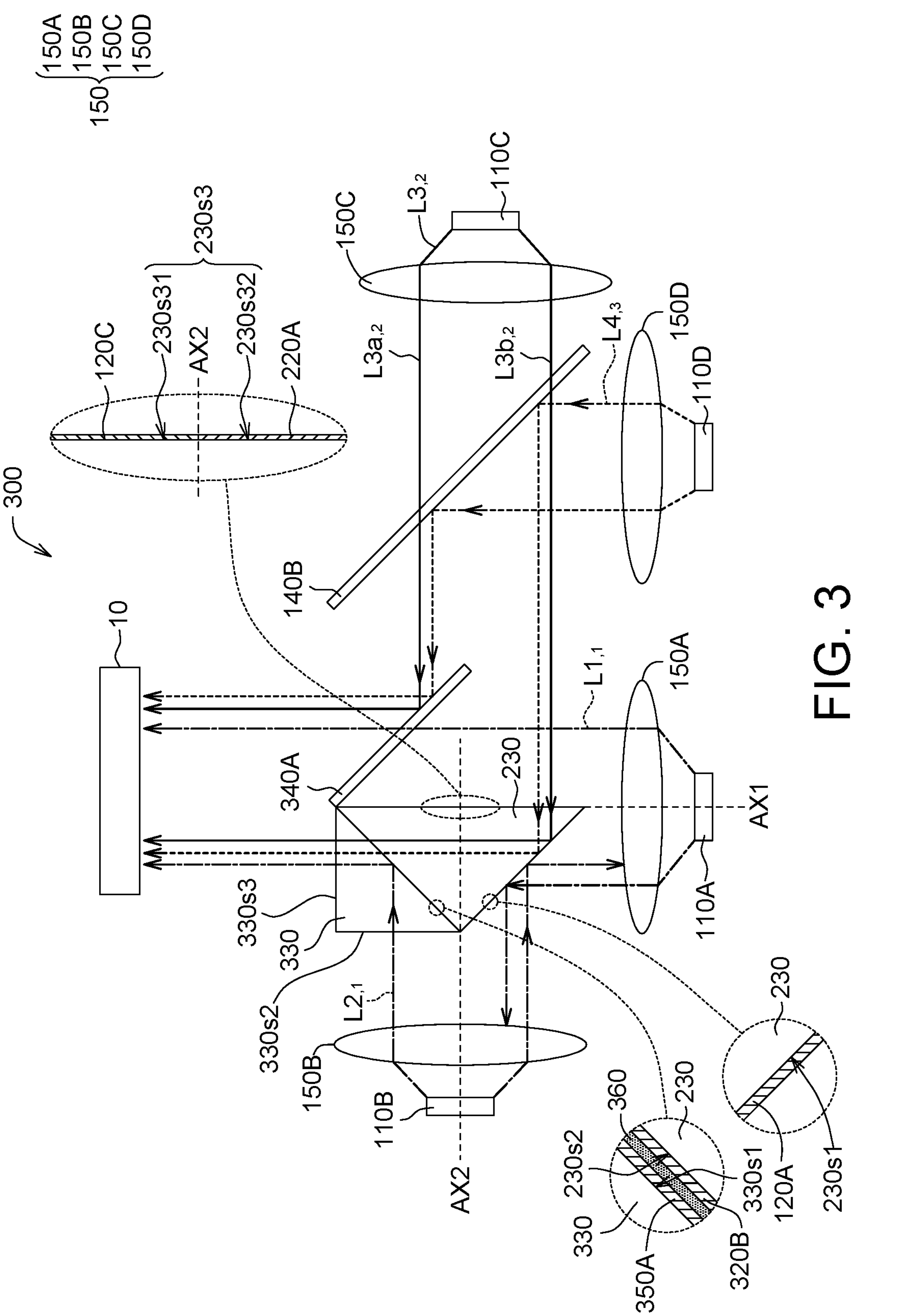
120B
230s2
230
120A
230s1

FIG. 2


150
150A
150B
150C
150D
300
10
120C
230s31
AX2
230s32
220A
230s3
150C
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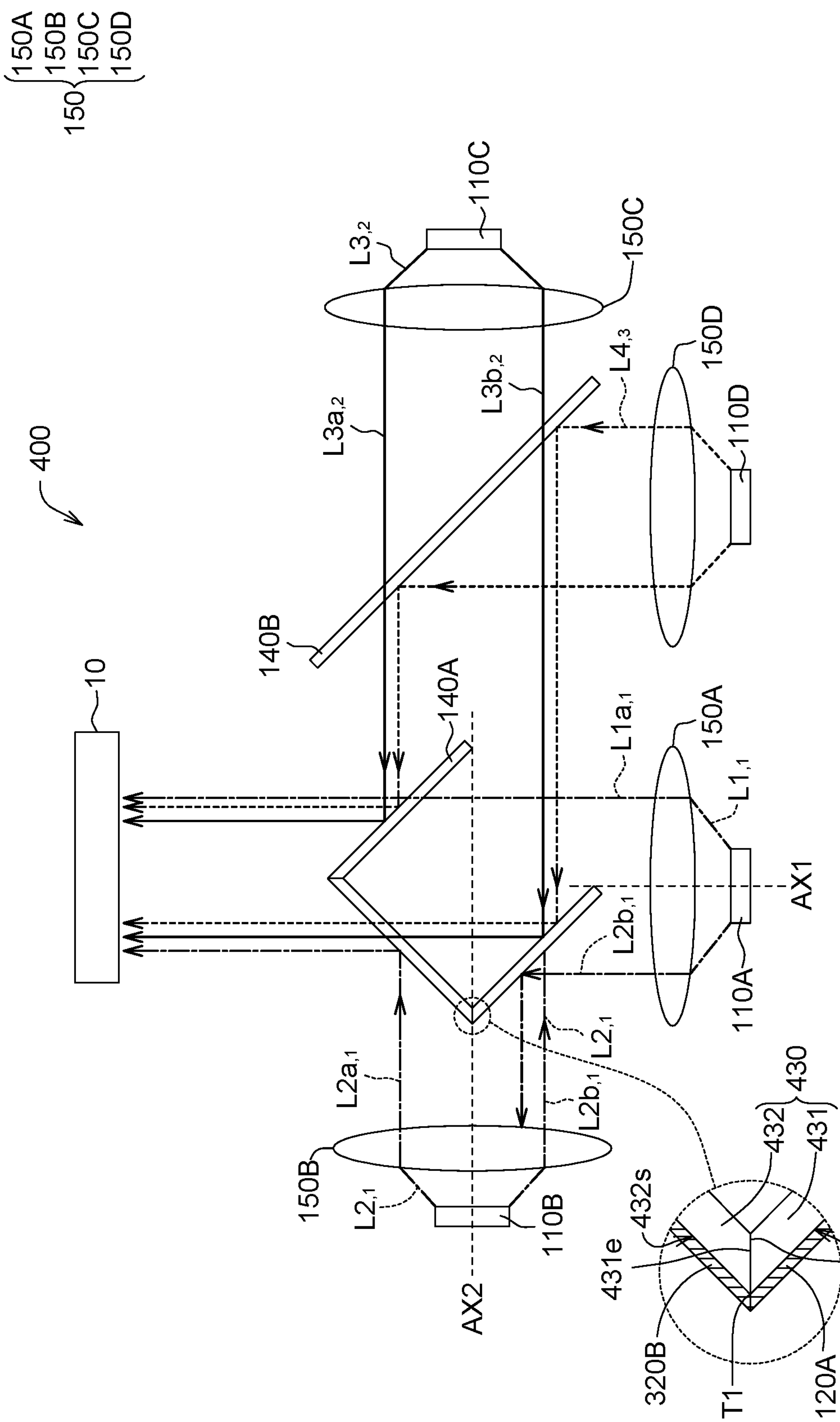
110C
L3a,2
L3b,2
L4,3
150D
110D
140B
340A
230
L1,1
150A
AX1
110A
330s3
330
330s2
L2,1
150B
110B
AX2
360
230
230s2
330s1
330
350A
320B
230
120A
230s1

FIG. 3



150
150A
150B
150C
150D
400
10
110A
110B
110C
110D
140A
140B
150A
150B
150C
150D
AX1
AX2
L1a,1
L1,1
L2a,1
L2b,1
L2,1
L3,2
L3a,2
L3b,2
L4,3
430
431
432
431s
432s
431e
432e
320B
T1
120A

FIG. 4

LIGHT SOURCE MODULE

This application claims the benefit of People's Republic of China application Serial No. 202211280226.2, filed on Oct. 19, 2022, the subject matter of which is incorporated herein by reference.

TECHNICAL FIELD

The disclosure relates in general to a light source module.

BACKGROUND

The light source module has a wide application, and many devices need the light source module, such as projectors, illuminator, flashlight, etc. Generally speaking, the optical components of the light source module may cause light loss due to factors such as assembly tolerances and spatial configurations. Therefore, proposing a new light source module capable of reducing light loss is one of the goals of the industry in this technical field.

SUMMARY

This disclosure proposes a light source module capable of improving the aforementioned conventional problems.

According to an embodiment of the present invention, a light source module is proposed. The light source module includes a first light source, a second light source, a first reflective layer and a second reflective layer. The first light source is configured to emit a first light having a first wavelength. The second light source is configured to emit a second light having the first wavelength. The first reflective layer is disposed opposite to the first light source and configured to reflect the first light. The second reflective layer is disposed opposite to the second light source and configured to reflect the second light. The first reflective layer and the second reflective layer are configured to reflect light of any wavelength in a visible spectrum.

According to another embodiment of the present invention, a light source module is provided. The light source module includes a first light source, a second light source, a light-transmitting element, a first reflective layer and a light-splitting layer. The first light source is configured to emit a first light having a first wavelength. The second light source is configured to emit a second light having the first wavelength. The light-transmitting element has a first surface and a second surface. The first reflective layer is formed on the first surface and configured to reflect the first light. The light-splitting layer is formed adjacent to the second surface and configured to reflect the second light.

The above and other aspects of the disclosure will become better understood with regard to the following detailed description of the preferred but non-limiting embodiment (s). The following description is made with reference to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** shows a schematic diagram of an optical path of a light source module **100** according to an embodiment of the present invention;

FIG. **2** shows a schematic diagram of an optical path of a light source module **200** according to another embodiment of the present invention;

FIG. **3** shows a schematic diagram of an optical path of a light source module **300** according to another embodiment of the present invention; and

FIG. **4** shows a schematic diagram of an optical path of a light source module **400** according to another embodiment of the present invention.

DETAILED DESCRIPTION

Referring to FIG. **1**, FIG. **1** shows a schematic diagram of an optical path of a light source module **100** according to an embodiment of the present invention. The light source module **100** could be applied to a device requiring a light source, such as a projector, illuminator, display or other types of devices. For application in the projector, the light source module **100** could also be called a light combining module.

As shown in FIG. **1**, the light source module **100** includes a first light source **110**A, a second light source **110**B, a first reflective layer **120**A, a second reflective layer **120**B, a third reflective layer **120**C, a first prism **130**, a reflective element **125**, a third light source **110**C, a fourth light source **110**D, a first light-splitting element **140**A, a second light-splitting element **140**B, at least one condenser lens **150** (for example, a first condenser lens **150**A, a second condenser lens **150**B, a third condenser lens **150**C and/or a fourth condenser lens **150**D).

As shown in FIG. **1**, the first light source **110**A is configured to emit a first light L**1**,$_1$ having a first wavelength (subscript "1" means "the first wavelength"). The second light source **1108** is configured to emit a second light L**2**,$_1$ having the first wavelength. The first reflective layer **120**A is disposed opposite to the first light source **110**A and configured to reflect the first light L**1**,$_1$. The second reflective layer **1208** is disposed opposite to the second light source **1108** and configured to reflect the second light L**2**,$_1$. The first reflective layer **120**A and the second reflective layer **1208** are configured to reflect light of any wavelength in the visible spectrum. As a result, the first reflective layer **120**A could reflect the first light L**1**,$_1$ back to the first light source **110**A, and the second reflective layer **1208** could reflect the second light L**2**,$_1$ back to the second light source **1108** to reuse the reflected light and reduce light loss.

As shown in FIG. **1**, the first light L**1**,$_1$ includes a first part L**1**a,$_1$ and a second part L**1**b,$_1$. The first part L**1**a,$_1$ and the second part L**1**b,$_1$ travel along two opposite sides of a first axis **AX1** respectively. The first axis **AX1** passes through a light-emitting surface of the first light source **110**A and the first condenser lens **150**A. For example, the first axis **AX1** passes through a center of the first light source **110**A and a center of the first condenser lens **150**A. The first part L**1**a,$_1$ could travel through the first light-splitting element **140**A, and the second part L**1**b,$_1$ is reflected from the first reflective layer **120**A back to the first light source **110**A for recycling.

As shown in FIG. **1**, the first light source **110**A further includes a first wavelength conversion layer **110A1**, a first reflective layer **110A2** and a first light-emitting layer **110A3**. The first light-emitting layer **110A3** is formed between the first wavelength conversion layer **110A1** and the first reflective layer **110A2**, and the first wavelength conversion layer **110A1** is closer to the second reflective layer **1208** than the first reflective layer **110A2**. The first light-emitting layer **110A3** includes, for example, at least one semiconductor epitaxial layer, which could emit a light L**1**'. In the present embodiment, the light L**1**' is, for example, light having the second wavelength or the third wavelength. The first wavelength conversion layer **110A1** includes several fluorescent particles **110**A**11**, which could excite the light to convert the wavelength of the light. For example, the light L**1'** having the second wavelength or the third wavelength is converted into the first light $L1,_1$ having the first wavelength. The first reflective layer **110**A**2** could reflect the light back to the first wavelength conversion layer **110**A**1** to increase the wavelength conversion efficiency. In the present embodiment, the first reflective layer **110**A**2** could reflect the second part $L1b,_1$, and the reflected second part $L1b,_1$ is divided into two parts traveling along opposite sides of the first axis AX**1** just like the aforementioned first light $L1,_1$.

As shown in FIG. **1**, the second light $L2,_1$ includes a third part $L2a,_1$ and a fourth part $L2b,_1$. The third part $L2a,_1$ and the fourth part $L2b,_1$ travel along two opposite sides of a second axis AX**2** respectively. The second axis AX**2** travels through a light-emitting surface of the second light source **1106** and the second condenser lens **150**B. For example, the second axis AX**2** passes through the center (for example, the optical axis) of the second light source **1106** and a center of the second condenser lens **150**B. The third part $L2a,_1$ could travel through the first light-splitting element **140**A, and the fourth part $L2b,_1$ is reflected from the reflective element **125** back to the second light source **1106** for recycling.

As shown in FIG. **1**, the second light source **1106** further includes a second wavelength conversion layer **110**B**1**, a second reflective layer **110**B**2** and a second light-emitting layer **110**B**3**. The second light-emitting layer **110**B**3** is formed between the second wavelength conversion layer **110**B**1** and the second reflective layer **110**B**2**, and the second wavelength conversion layer **110**B**1** is closer to the reflective element **125** than the second reflective layer **110**B**2**. The second light-emitting layer **110**B**3**, for example, includes at least one semiconductor epitaxial layer, which could emit a light L**2'**. In the present embodiment, the light L**2'** is light having the second wavelength or the third wavelength, for example. The second wavelength conversion layer **110**B**1** includes several fluorescent particles **110**B**11** which could excite the light to convert the wavelength of the light. For example, the light L**2'** having the second wavelength or the third wavelength is converted into the second light $L2,_1$ having the first wavelength. The second reflective layer **110**B**2** could reflect the light back to the second wavelength conversion layer **110**B**1** to increase the wavelength conversion efficiency. In the present embodiment, the second reflective layer **110**B**2** could reflect the fourth part $L2b,_1$, and the reflected fourth part $L2b,_1$ is divided into two parts traveling along two opposite sides of the second axis AX**2** just like the aforementioned second light $L2,_1$.

As shown in FIG. **1**, the third light source **110**C is configured to emit a third light $L3,_2$ having a second wavelength (subscript "2" means "second wavelength"). The third light $L3,_2$ could travel through the second light-splitting element **140**B, be incident to the first light splitting element **140**A, and then be reflected from the first light splitting element **140**A to be incident to the module **10**. The module **10** is, for example, an illumination module or an imaging module or light pipe.

As shown in FIG. **1**, the fourth light source **110**DC is configured to emit a fourth light $L4,_3$ having a third wavelength (subscript "3" means "third wavelength"). The fourth light $L4,_3$ is incident to the second light-splitting element **140**B, reflected from the second light-splitting element **140**B to the first light-splitting element **140**A, and then reflected from the first light-splitting element **140**A to be incident the module **10**.

Light of different wavelengths has different light colors. In terms of light color, the aforementioned color light with the first wavelength is, for example, one of blue light (for example, the wavelength is between 450 nm and 495 nm), red light (for example, the wavelength is between 620 nm and 750 nm) and green light (The wavelength is, for example, one of 495 nm to 570 nm), wherein the color light having the second wavelength is, for example, another of the blue light, the red light and the green light, and the color light having the third wavelength is, for example, the other of the blue light, the red light and the blue light. In the present embodiment of the present invention, the color light having the first wavelength is the green light, the color light having the second wavelength is the red light, and the color light having the third wavelength is the blue light.

As shown in FIG. **1**, the first reflective layer **120**A and the second reflective layer **120**B could reflect light of any wavelength in the visible spectrum. In other words, the first reflective layer **120**A and the second reflective layer **120**B do not allow the light of any wavelength in the visible spectrum to travel through. The first reflective layer **120**A and the second reflective layer **120**B are, for example, reflective mirrors, or the first reflective layer **120**A and the second reflective layer **120**B may exclude the light-splitting element.

As shown in FIG. **1**, the third reflective layer **120**C could reflect light of any wavelength in the visible spectrum. In other words, the third reflective layer **120**C does not allow the light of any wavelength in the visible spectrum to travel through. The third reflective layer **120**C is, for example, a reflective mirror, or the third reflective layer **120**C may exclude a light-splitting element. The third reflective layer **120**C could reflect the first part $L1a,_1$ of the first light $L1,_1$ (if the first part $L1a,_1$ is incident on the third reflective layer **120**C).

As shown in FIG. **1**, the reflective element **125** is disposed opposite to the second light source **1108**. The reflective element **125** is disposed on a side of the second axis AX**2** for reflecting the third part $L2a,_1$. In the present embodiment, the reflective element **125** may be a light diffraction (or refractive) element, such as a reflective mirror, a light-splitting element or the like. The reflective element **125** and the first prism **130** are respectively disposed on two opposite sides of the second axis AX**2**.

As shown in FIG. **1**, the first prism **130** is disposed on a side of the first axis AX**1** and/or on a side of the second axis AX**2**. The first reflective layer **120**A and the second reflective layer **1208** are formed on the first prism **130**. For example, the first prism **130** has a first surface **130***s***1** and a second surface **130***s***2**. The first reflective layer **120**A is formed on the first surface **130***s***1**, and the second reflective layer **1208** is formed on the second surface **130***s***2**. The first prism **130** has a polygonal cross section, such as a triangle. The first surface **130***s***1** and the second surface **130***s***2** are two adjacent or connecting surfaces of the first prism **130** respectively. In an embodiment, an included angle between the first surface **130***s***1** and the second surface **130***s***2** is, for example, 45 degrees, but it could also be greater or smaller.

As shown in FIG. **1**, since the first prism **130**, the first reflective layer **120**A, the second reflective layer **1208** and the third reflective layer **120**C form an integral structure which could move together, it is convenient to assemble in the light path space within the light source module **100**. In addition, the first reflective layer **120**A and the second reflective layer **120**B could be connected to each other at an junction T**1** between the first surface **130***s***1** and the second surface **130***s***2** of the first prism **130**, and thus it could prevent a physical light-transmitting material of the first prism **130** from be exposed from the junction T**1**, thereby preventing the second light $L2,_1$ from traveling the first prism **130** to become light leakage (light loss). Similarly, the second reflective layer **120**B and the third reflective layer **120**C could be connected to each other at a junction T**2** between the second surface **130***s***2** and a third surface **130***s***3** of the first prism **130**, and thus it could prevent the physical light-transmitting material of the first prism **130** from be exposed from the junction T**2**, thereby reducing or preventing the (light leakage). Similarly, the first reflective layer **120**A and the third reflective layer **120**C could be connected to each other at a junction T**3** between the first surface **130***s***1** and the third surface **130***s***3** of the first prism **130**, and thus it could prevent the physical light-transmitting material of the first prism **130** from be exposed from the junction T**3**, thereby reducing or preventing the (light leakage).

As shown in FIG. **1**, the third reflective layer **120**C is formed on the first prism **130**. For example, the first prism **130** further has the third surface **130***s***3**. The third reflective layer **120**C is formed on the third surface **130***s***3**. The first surface **130***s***1** and the third surface **130***s***3** are two adjacent or connecting surfaces of the first prism **130** respectively. In an embodiment, an included angle between the first surface **130***s***1** and the third surface **130***s***3** is, for example, 90 degrees, but it could also be greater or smaller.

As shown in FIG. **1**, the first reflective layer **120**A, the second reflective layer **120**B and/or the third reflective layer **120**C could be formed or disposed on at least one lateral surface of the first prism **130**. In detail, the first reflective layer **120**A, the second reflective layer **120**B and/or the third reflective layer **120**C are, for example, coatings coated on the first prism **130**, or reflective films, reflective sheets or reflective mirrors attached to the first prism **130**.

As shown in FIG. **1**, the first light-splitting element **140**A is, for example, a dichroic beam splitter. The first light-splitting element **140**A is disposed opposite to the first light source **110**A, the second light source **1108** and the third light source **110**C. The first light-splitting element **140**A could reflect the light having the second wavelength and the light having the third wavelength, but allow the light having the first wavelength to travel through. For example, the first light-splitting element **140**A could reflect the third light $L3,_2$ and the fourth light $L4,_3$ but allow the first light $L1,_1$ and the second light $L2,_1$ to travel through. A mixed light of the first light $L1,_1$, the second light $L2,_1$, the third light $L3,_2$ and the fourth light $L4,_3$ traveling through the first light-splitting element **140**A is, for example, white light.

In an embodiment, the first light $L1,_1$ and the second light $L2,_1$ are, for example, green light. Green light accounts for about 70% of white light, and the higher the proportion of green light, the higher the brightness of white light. Since the light source module **100** provides the green light with a high light-flux/a high brightness, the brightness of the white light emitted by the light source module **100** could be enhanced.

As shown in FIG. **1**, the second light-splitting element **140**B is, for example, a dichroic beam splitter. The second light-splitting element **140**B is disposed opposite to the third light source **110**C and the fourth light source **110**D. The second light-splitting element **140**B could reflect the light having the third wavelength, but allow the light having the second wavelength to travel through. For example, the second light-splitting element **140**B could reflect the fourth light $L4,_3$ but allow the third light $L3,_2$ to travel through.

The condenser lens could condense the light emitted by the light source, so that the light traveling through the condenser lens becomes a collimated light. The condenser lens includes at least one lens which could be a spherical lens, an aspheric lens or a combination thereof.

As shown in FIG. **1**, the first condenser lens **150**A is disposed opposite to the first light source **110**A to collimate the first light $L1,_1$. The first condenser lens **150**A is disposed in the first axis AX**1**. For example, the first axis AX**1** passes through the center of the first condenser lens **150**A, so that the first part $L1a,_1$ and the second part $L1b,_1$ incident on the first condenser lens **150**A have the substantial same amount of light relative to the first axis AX**1**.

As shown in FIG. **1**, the second condenser lens **1506** is disposed opposite to the second light source **1106** to collimate the second light $L2,_1$. The second condenser lens **1506** is disposed in the second axis AX**2**. For example, the second axis AX**2** passes through the center of the second condenser lens **150**B, so that the third part $L2a,_1$ and the fourth part $L2b,_1$ incident to the second condenser lens **150**B have the substantial same amount of light relative to the first axis AX**1**.

As shown in FIG. **1**, the third condenser lens **150**C is disposed opposite to the third light source **110**C to collimate the third light $L3,_2$. The fourth condenser lens **150**D is disposed opposite to the fourth light source **110**D to collimate the fourth light $L4,_3$.

Referring to FIG. **2**, FIG. **2** shows a schematic diagram of an optical path of a light source module **200** according to another embodiment of the present invention. The light source module **200** could be applied to a device requiring a light source, such as a projector, illuminator, display or other types of devices. For application in the projector, the light source module **200** could also be called a light combining module.

As shown in FIG. **2**, the light source module **200** includes the first light source **110**A, the second light source **1106**, the first reflective layer **120**A, the second reflective layer **120**B, the third reflective layer **120**C, a first light-transmitting layer **220**A, a first prism **230**, the third light source **110**C, the fourth light source **110**D, the first light-splitting element **140**A, the second light-splitting element **140**B, at least one condenser lens **150** (for example, the first condenser lens **150**A, the second condenser lens **150**B, the third condenser lens **150**C and/or or the fourth condenser lens **150**D).

The light source module **200** of the present embodiment includes the structure the same as or similar to that of the aforementioned light source module **100**, and difference is that, for example, the first prism **230** of the light source module **200** and the first prism **130** of the light source module **100** are different in structure.

As shown in FIG. **2**, the first prism **230** has a first surface **230***s***1**, the second surface **230***s***2** and a third surface **230***s***3**. The first surface **230***s***1** is connected adjacent to the second surface **230***s***2**, and an angle between the first surface **230***s***1** and the second surface **230***s***2** is, for example, 90 degrees, but it could also be greater or smaller. The first surface **230***s***1** is connected adjacent to the third surface **230***s***3**, and an angle between the first surface **230***s***1** and the third surface **230***s***3** is, for example, 45 degrees, but it could also be greater or smaller. The second surface **230***s***2** is connected adjacent to the third surface **230***s***3**, and an angle between the second surface **230***s***2** and the third surface **230***s***3** is, for example, 45 degrees, but it could also be greater or smaller.

As shown in FIG. **2**, the first surface **230***s***1** and the second surface **230***s***2** are respectively located on opposite two sides of the second axis AX**2**. As a result, the first surface **230***s***1** could reflect the fourth part $L2b,_1$ of the second light $L2,_1$, and the second surface **230***s***2** could reflect the third part $L2a,_1$ of the second light $L2,_1$. In an embodiment, a projected area of the first surface **230***s***1** and the second surface **230***s***2** along the second axis AX**2** onto the second condenser lens **150**B completely (fully) overlaps with the second condenser lens **150**B, and thus it could reflect the entire of the second light $L2_{,1}$ traveling through the second condenser lens **1508** to reduce or avoid light loss (light leakage). In addition, since the projected area of the first surface **230**s**1** and the second surface **230**s**2** along the second axis AX**2** onto the second condenser lens **150**B completely (fully) overlaps with the second condenser lens **150**B, even if there is an assembly tolerance between the first prism **230** and other elements (for example, the second light source **110**B), it will not cause light leakage (light loss). Furthermore, since the projected area of the first surface **230**s**1** and the second surface **230**s**2** along the second axis AX**2** onto the second condenser lens **150**B completely (fully) overlaps with the second condenser lens **150**B, an assembly tolerance or a positional deviation (for example, a tolerance resulted from the assembly jig, a positional deviation of supporting points in the light source module, manual errors of assemblers, etc.) between the first prism **230** and other elements (for example, the second light source **1108**) is allowed.

As shown in FIG. **2**, the third surface **230**s**3** includes a first surface portion **230**s**31** and a second surface portion **230**s**32**. The second axis AX**2**, for example, passes through a boundary between the first surface portion **230**s**31** and the second surface portion **230**s**32**, that is, the first surface portion **230**s**31** and the second surface portion **230**s**32** are respectively located on two opposite sides of the second axis AX**2**. The third reflective layer **120**C is formed on the first surface portion **230**s**31** of the third surface **230**s**3**. The first light-transmitting layer **220**A is formed on the second surface portion **230**s**32** of the third surface **230**s**3**. Compared with the second surface portion **230**s**32**, the probability or proportion of the first part $L1a_{,1}$ of the first light $L1_{,1}$ incident on the first surface portion **230**s**31** is higher, and thus the third reflective layer **120**C may be formed on the first surface portion **230**s**31** for reflecting the first part $L1a_{,1}$, while the first light-transmitting layer **220**A or a reflective layer (for example, another third reflective layer **120**C) may be optionally formed on the second surface portion **230**s**32**.

As shown in FIG. **2**, since the prism **230**, the first reflective layer **120**A, the second reflective layer **1208**, the third reflective layer **120**C and the first light-transmitting layer **220**A form an integral structure, and thus they could move together. As a result, it is convenient to be assembled in the light path space of the light source module **200**. In addition, the first reflective layer **120**A and the second reflective layer **1208** could be connected to each other at the junction T**1** between the first surface **230**s**1** and the second surface **230**s**2** of the first prism **230**, and thus it could prevent a physical light-transmitting material of the first prism **230** from be exposed from the junction T**1**, thereby preventing the second light $L2_{,1}$ from traveling the first prism **230** to become light leakage (light loss). Similarly, the second reflective layer **1208** and the third reflective layer **120**C could be connected to each other at the junction T**2** between the second surface **230**s**2** and the third surface **230**s**3** of the first prism **230**, and thus it could prevent the physical light-transmitting material of the first prism **230** from be exposed from the junction T**2**, thereby reducing or preventing the (light leakage). Similarly, the first reflective layer **120**A and the first light-transmitting layer **220**A could be connected to each other at the junction T**3** between the first surface **230**s**1** and the third surface **230**s**3** of the first prism **230**, and thus it could prevent the physical light-transmitting material of the first prism **230** from be exposed from the junction T**3**, thereby reducing or preventing the (light leakage). The method of forming the first light-transmitting layer **220**A on the prism may be the same as or similar to that of the first reflective layer **120**A, the second reflective layer **120**B, and/or the third reflective layer **120**C, and it will not be repeated here.

Referring to FIG. **3**, FIG. **3** shows a schematic diagram of an optical path of a light source module **300** according to another embodiment of the present invention. The light source module **300** could be applied to a device requiring a light source, such as a projector, illuminator, display or other types of devices. For application in the projector, the light source module **300** could also be called a light combining module.

As shown in FIG. **3**, the light source module **300** includes the first light source **110**A, the second light source **1108**, the third light source **110**C, the fourth light source **110**D, the first reflective layer **120**A, the second reflective layer **120**B, the third reflective layer **120**C, the second light-splitting element **140**B, at least one condenser lens **150** (for example, the first condenser lens **150**A, the second condenser lens **150**B, the third condenser lens **150**C and/or the fourth condenser lens **150**D), the first light-transmitting layer **220**A, the first prism **230** (light-transmitting element), a light-splitting layer **320**B, anti-reflective layer **350**A, a second prism **330**, a first light-splitting element **340**A and an adhesive (or glue) layer **360**.

As shown in FIG. **3**, the first light source **110**A is configured to emit the first light $L1_{,1}$ having the first wavelength. The second light source **1106** is configured to emit the second light $L2_{,1}$ having the first wavelength. The first prism **230** has the first surface **230**s**1** and the second surface **230**s**2**. The first reflective layer **120**A is formed on the first surface **230**s**1** and is configured to reflect the first light $L1_{,1}$. The light-splitting layer **320**B is formed adjacent to the second surface **230**s**2** and configured to reflect the second light $L2_{,1}$. In addition, compared with the light source module **200**, the second light-splitting element **1406** and the third light source **110**C of the light source module **300** are located closer to the fourth light source **110**D, and thus it could shorten the length of the optical path of the fourth light $L4_{,3}$.

As shown in FIG. **3**, the light-splitting layer **320**B allows the light having the second wavelength and the light having the third wavelength to travel through. For example, the light-splitting layer **320**B allows the third light $L3_{,2}$ and the fourth light $L4_{,3}$ to travel through. The light-splitting layer **320**B is formed between two opposite surfaces of the first prism **130** and the second prism **330**. For example, the second prism **330** has a fourth surface **330**s**1**, the fourth surface **330**s**1** of the second prism **330** is opposite to the second surface **230**s**2** of the first prism **230**, and the light-splitting layer **320**B could be pre-formed on the fourth surface **330**s**1** of the second prism **330** or pre-formed on the second surface **230**s**2** of the first prism **230**.

As shown in FIG. **3**, the second prism **330** is connected adjacent to the first prism **230**. For example, the adhesive layer **360** may be formed between the fourth surface **330**s**1** of the second prism **330** and the second surface **230**s**2** of the first prism **230** to bond the second prism **330** and the first prism **230**. There is no air layer in the space between the two opposite surfaces of the second prism **330** and the first prism **230**, and thus it could avoid light loss caused by excessive deflection of light. For example, space between the fourth surface **330**s**1** of the second prism **330** and the second surface **230**s**2** of the first prism **230** is filled with the light-splitting layer **320**B, the adhesive layer **360** and the anti-reflective layer **350**A. In an embodiment, the first prism **230** has a first refraction index, the second prism **330** has a second refraction index, the adhesive layer **360** has a third refraction index, wherein the first refraction index, the second refraction index and the third refraction index are substantially equal to avoid light loss caused by excessive deflection of light.

As shown in FIG. **3**, the anti-reflective layer **350**A could be formed on the fourth surface **330**_s_**1** of the second prism **330**. The anti-reflective layer **350**A has a light transmittance of at least 95%. The method of forming the anti-reflective layer **350**A on the prism may be the same as or similar to that of the first reflective layer **120**A, the second reflective layer **120**B and/or the third reflective layer **120**C, and it will not be repeated here.

As shown in FIG. **3**, the second prism **330** further has a fifth surface **330**_s_**2** and a sixth surface **330**_s_**3**, wherein the fifth surface **330**_s_**2** is adjacent to or connected to the sixth surface **330**_s_**3**, and an included angle between the fifth surface **330**_s_**2** and the sixth surface **330**_s_**3** is, for example, 90° degree, but could also be greater or smaller. Although not shown, the light source module **300** further includes two anti-reflective layers, and the two anti-reflective layers are formed on the fifth surface **330**_s_**2** and the sixth surface **330**_s_**3** respectively.

As shown in FIG. **3**, in the present embodiment, the second axis AX**2** further passes through a light-emitting surface of the third light source **110**C. The third light $L3,_2$ includes a fifth part $L3a,_2$ and a sixth part $L3b,_2$. The fifth part $L3a,_2$ and the sixth part $L3b,_2$ travel along two opposite sides of the second axis AX**2** respectively. The first light-splitting element **340**A is disposed on a side of the second axis AX**2**. The fifth part $L3a,_2$ is incident to the first light-splitting element **340**A after traveling through the second light-splitting element **1408**, and then reflects from the first light-splitting element **340**A to the module **10**. The sixth part $L3b,_2$ is incident to the first reflective layer **120**A after traveling through the second light-splitting element **140**B, and then reflects from the first reflective layer **120**A to the module **10**.

Referring to FIG. **4**, FIG. **4** shows a schematic diagram of an optical path of a light source module **400** according to another embodiment of the present invention. The light source module **400** could be applied to a device requiring a light source, such as a projector, illuminator, display or other types of devices. For application in the projector, the light source module **400** could also be called a light combining module.

As shown in FIG. **4**, the light source module **400** includes the first light source **110**A, the second light source **1108**, the first reflective layer **120**A, a light-transmitting element **430**, the third light source **110**C, the fourth light source **110**D and the first light-splitting element **140**A, the second light-splitting element **140**B, at least one condenser lens **150** (for example, the first condenser lens **150**A, the second condenser lens **150**B, the third condenser lens **150**C and/or the fourth condenser lens **150**D) and the light-splitting layer **320**B.

As shown in FIG. **4**, the light-transmitting element **430** includes a first light-transmitting plate **431** and a second light-transmitting plate **432**, wherein the first light-transmitting plate **431** has a first surface **431**_s_, the second light-transmitting plate **432** has a second surface **432**_s_, and an angle between the first light-transmitting plate **431** and the second light-transmitting plate **432** is, for example, 90 degrees, but it could also be greater or smaller. In addition, the first light-transmitting plate **431** and the second light-transmitting plate **432** may be integrally formed, or connected together after formed separately.

As shown in FIG. **4**, the first light-transmitting plate **431** has an end surface **431**_e_, and the second light-transmitting plate **432** has an end surface **432**_e_, wherein the end surface **431**_e_ is connected to the end surface **432**_e_. The first reflective layer **120**A is formed on the first surface **431**_s_ of the first light-transmitting plate **431**. The light-splitting layer **320**B is formed on the second surface **432**_s_ of the second light-transmitting plate **432**. In the present embodiment, the first reflective layer **120**A and the light-splitting layer **320**B are connected at a junction T**1** between the first surface **431**_s_ and the second surface **432**_s_, and thus it could prevent a physical light-transmitting material of the first light-transmitting plate **431** and a physical light-transmitting material of the second light-transmitting plate **432** from be exposed from the junction T**1**, thereby reducing or preventing the (light leakage). In addition, the second light-transmitting plate **432** and the first light-splitting element **140**A have the connection features similar to or the same as that the first light-transmitting plate **431** and the second light-transmitting plate **432**, and it will not be repeated here. The light-splitting layer **320**B could extend to the junction T**2** between the second light-transmitting plate **432** and the first light-splitting element **140**A, and it could reduce or avoid light loss (light leakage).

As shown in FIG. **4**, the light-transmitting element **430**, the first reflective layer **120**A and the light-splitting layer **320**B could be an integral structure, and thus they could move together. As a result, it is convenient to be assembled in the light path space of the light source module **400**. In an embodiment, the light-transmitting element **430**, the first reflective layer **120**A, the light-splitting layer **320**B and the first light-splitting element **140**A could be an integral structure, and thus they could move together. As a result, it is convenient to be assembled in the light path space of the light source module **400**.

To sum up, the embodiment of the present invention proposes a light source module. In an embodiment, the light source module includes at least one light source and at least one reflective layer. Each light source could emit light of a first wavelength, and the reflective layer could reflect the light of the first wavelength back to the light source to recycling and reuse for the light of the wavelength, thereby reducing light loss and/or enhancing the brightness of light output.

It will be apparent to those skilled in the art that various modifications and variations could be made to the disclosed embodiments. It is intended that the specification and examples be considered as exemplary only, with a true scope of the disclosure being indicated by the following claims and their equivalents.

What is claimed is:

1. A light source module, comprising: a first light source configured to emit a first light having a first wavelength; a second light source configured to emit a second light having the first wavelength; a first reflective layer disposed opposite to the first light source and configured to reflect the first light; and a second reflective layer disposed opposite to the second light source and configured to reflect the second light; the first reflective layer and the second reflective layer are configured to reflect light of any wavelength in a visible spectrum; a first prism having a first surface and a second surface; the first reflective layer is formed on the first surface, and the second reflective layer is formed on the second surface and a reflective element disposed opposite to the second light source; wherein the first prism and the reflective element are disposed on two opposite sides of an axis respectively, and the axis passes through a light-emitting surface of the second light source.

2. The light source module as claimed in claim **1**, wherein an angle between the first surface and the second surface is 45 degrees.

3. The light source module as claimed in claim **1**, wherein the first prism is disposed on a side of an axis, and the axis passes through a light-emitting surface of the first light source.

4. The light source module as claimed in claim **1**, further comprising:

a third light source configured to emit a third light with a second wavelength;

a first light-splitting element disposed opposite to the first light source, the second light source and the third light source;

wherein the first light-splitting element is configured to reflect the third light but allow the first light and the second light to travel through.

5. The light source module as claimed in claim **1**, wherein an angle between the first surface and the second surface is 90 degrees.

6. The light source module as claimed in claim **4** further comprising:

a fourth light source configured to emit a fourth light having a third wavelength;

a second light-splitting element disposed opposite to the third light source and the fourth light source;

wherein the second light-splitting element is configured to reflect the fourth light but allow the third light to travel through.

7. The light source module as claimed in claim **5**, further comprising:

a fourth light source configured to emit a fourth light having a third wavelength;

a second light-splitting element disposed opposite to the third light source and the fourth light source;

wherein the second light-splitting element is configured to reflect the fourth light but allow the third light to travel through.

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